

YUHENG LU(卢昱恒)

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EDUCATION BACKGROUND

Harbin Institute of Technology (HIT), China

2018.09 – 2022.07

BEng in Electrical Engineering & Automation

Overall GPA: **91.15**/100, Major GPA: **90.73**/100, Rank: **13/296(top 5%)**

École Polytechnique Fédérale de Lausanne(EPFL), Switzerland

2022.09 – 2025.07

Master of Science in Electrical and Electronics Engineering, Minor in Computer Science

Overall GPA: **4.93**/6.0

Research field: Electrical Engineering

Waseda University, Japan

2025.09 – present

Master of Engineering in Information in Graduate School of Information, Production and Systems

Overall GPA: **3.13**/4

Research field: Deep Research Agent, Information Retrieval, Remote Sensing Agent

RESEARCH

[2026 ACL Findings] Code-Switching Information Retrieval: Benchmarks, Analysis, and the Limits of Current Retrievers

2025.09 – 2026.01

Co-first Author; [Arxiv Link](#); [Github Repository Link](#)

- Built two code-switching retrieval benchmarks: CSR-L (human-annotated, 4 IR datasets, En–Zh / En–Ja) and CS-MTEB (LLM-generated, 11 tasks / 7 categories / 9 language pairs).
- Benchmarked 4 retriever families (BM25, bi-encoder, cross-encoder, ColBERT v2), revealing consistent code-switching degradation of up to 27% even for strong multilingual models.
- Visualized the failure via PCA embedding-space, showing original vs. code-switched queries form disjoint clusters, evidencing the problem is representational, not surface-level.
- Proposed a lexicon-based vocabulary expansion method that partially closed the gap.

[Preprint] Beyond Monolingual Deep Research: Evaluating Agents and Retrievers with Cross-Lingual BrowseComp-Plus

2026.01 – 2026.05

Co-first Author; [Arxiv Link](#); [Github Repository Link](#)

- Built XBCP, the first benchmark for cross-lingual deep research, extending BrowseComp-Plus into two settings: cross-lingual (single-language evidence per query) and multilingual (evidence distributed equally across 12 high/low-resource languages) by translating the evidence corpus via GPT-5.4 with external expert quality verification.
- Evaluated 4 deep research agents $\times$ 5 retrievers across 3 corpus conditions, measuring accuracy, evidence recall, calibration, and citation fidelity.
- Revealed large cross-lingual degradation: accuracy drops $\sim$ 16–23 pp with translated evidence even for strong dense retrievers and used oracle retrieval to isolate a persistent agent-side integration bottleneck beyond retrieval failure.
- Ran diagnostic analyses (per-language decomposition, reasoning-based query expansion +3.25 pp, reasoning-effort control), further decomposing the agent bottleneck into evidence-understanding and prompt-language penalties.

[Under Review] Social-BRIGHT: Benchmarking Socially Grounded Reasoning in Information Retrieval

2026.01 – 2026.05

Second Author; Arxiv Link and Github Repository coming soon

- Built Social-BRIGHT, a benchmark casting 4 forms of social interpretation (interpersonal response, social group grounding, humor explanation, dogwhistle) into full-corpus retrieval with 2,154 queries and matched pairs, each pairing a socially correct document with a lexically plausible but misleading hard negative, curated via GPT-5.4 synthesis and human validation.
- Introduced a matched-pair accuracy (MPA) metric to isolate social discrimination from recall, and evaluated 18+ retrievers across bi-encoder and late-interaction families zero-shot under a unified MTEB protocol.
- Showed current retrievers fail at socially grounded relevance, proving the bottleneck is fine-grained discrimination rather than retrieval.
- Analyzed 3 enhancements (reasoning retrievers, reranking, query expansion) with per-subset and query-shuffle controls, finding selective gains that do not close the gap.

[Ongoing] Agentic Orchestration of Single-Degradation Experts for Compound Remote Sensing Restoration

2026.05 - present

Second Author; Arxiv Link and Github Repository coming soon

- Proposed an LLM-orchestrated agent that decomposes compound degradations and composes single-degradation expert tools step-by-step, outperforming all-in-one models on remote sensing restoration.
- Designed a rollback-guarded step-wise execution loop that keeps only quality-improving steps (PSNR/SSIM/LPIPS), guaranteeing monotonic improvement and suppressing error accumulation across the tool.
- Building a self-maintaining tool library with automatic discovery (GitHub/arXiv search agent) and verification-based retirement, pruning old tools to mitigate library drift.(still ongoing)

INTERNSHIP

Autonomous Driving Algorithm Research Engineer

2024.07 – 2024.12

SenseTime Research, Beijing China

--Project UniAD

- Developed a data pipeline for collecting, cleaning, extracting and visualizing data(ETL)

--Project DriveAGI

- Used GPT-4o's API to create QA pairs as data source for training large language model that recognizes car brands and types employed on vehicles.
- Designed an efficient and easy-to-modify data pipeline for processing datasets returned by GPT 4o's API.

SKILLS

Programming Languages: Python as main programming language; SQL used in my internship

English Grades: IELTS 7.5(6.5), GRE 325+4.0; **Japanese Grade:** JLPT N2.

HONORS AND AWARDS

- National Scholarship, the Ministry of Education of China 2018 - 2019
- Infineon Technology Scholarship, Infineon China 2019 - 2020
- Outstanding Student, Harbin Institute of Technology 2019 - 2020